

CardioPredictor: A Real-time Non-Invasive and Cuffless Cardiovascular Disease Prediction System using 10-Fold Cross Validation on The Framingham Dataset

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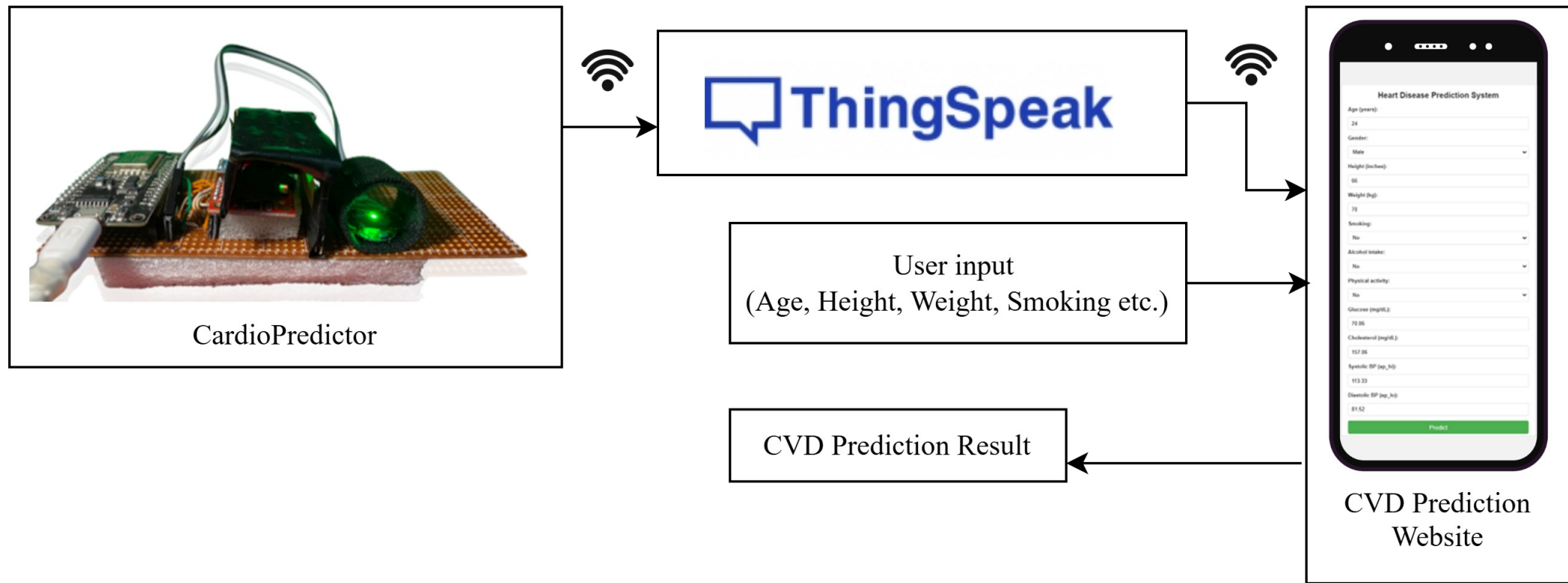


Fig. 1: System block diagram of the overall CardioPredictor system

Invasive Vs Non-Invasive Method



Fig 2: Invasive method of measuring glucose

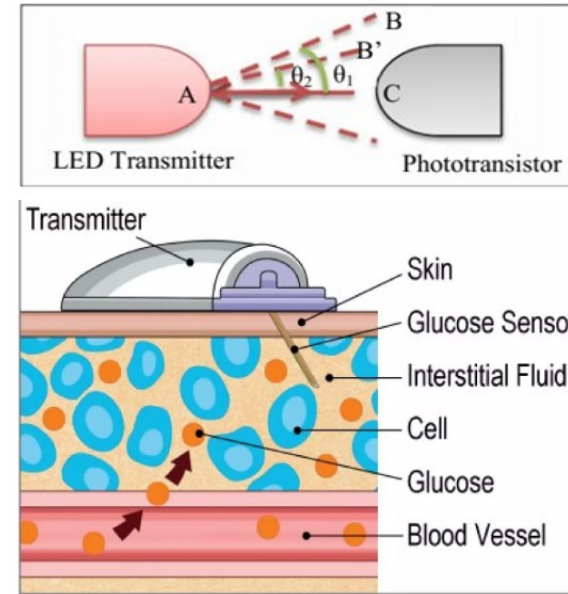


Fig 3: Non-Invasive method of measuring glucose

$$\text{Glucose (mg/dL)} = -0.0011X + 194.87 \dots \dots \dots (\text{eq 1})'$$
$$\text{Cholesterol (mg/dL)} = -0.0033X + 531.49 \dots \dots \dots (\text{eq 2})$$

here, X is the average value of IR

Cuff-based Vs Cuffless Method



Fig 4: Blood Pressure measurement with Cuffbased system

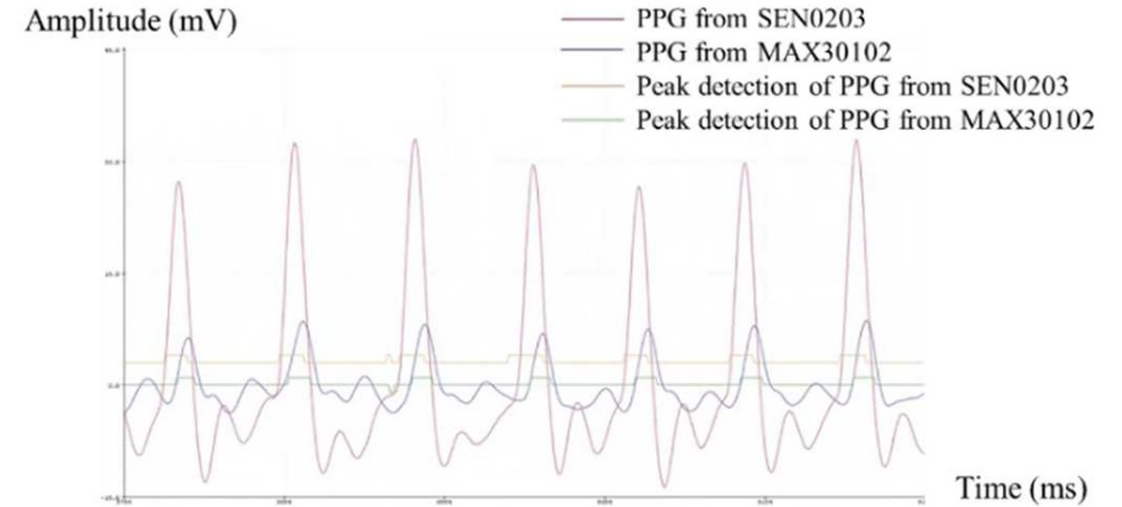


Fig 5: Blood Pressure measurement with Cuffless system (PTT Method)

$$\text{SBP (mmHg)} = 184.3 - 1.329 \times \text{HR(bpm)} + 0.00848 \times Td \dots\dots\dots(\text{eq 3})$$

$$\text{DBP (mmHg)} = 55.96 - 0.02912 \times \text{HR(bpm)} + 0.02302 \times Td \dots\dots\dots(\text{eq 4})^2$$

here, $Td = \text{HR(ms)} - \text{Time Delay}$; $\text{Time Delay} = \text{HR(bpm)} + \text{PTT}$

Circuit Diagram

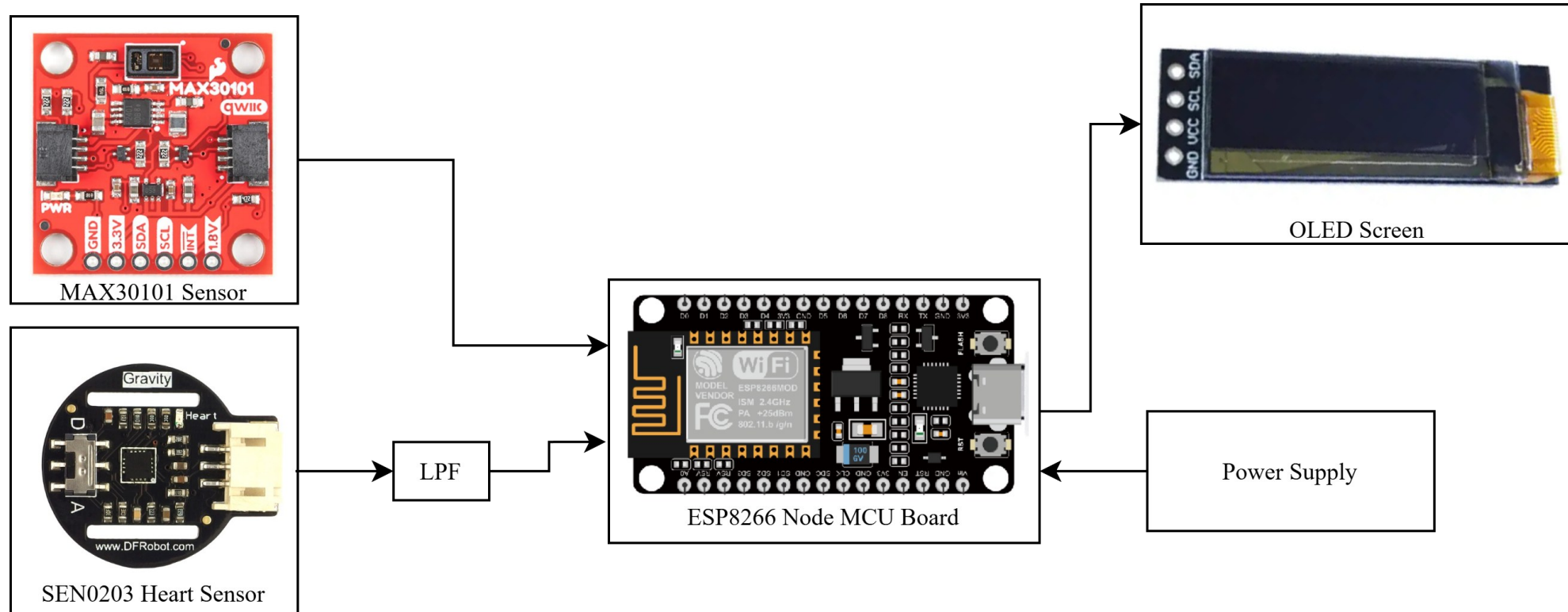


Fig. 6: Circuit diagram CardioPredictor

Methodology

Table 1: Required data for the model

SL No.	Data	Specification
01	Glucose Level	mg/dl
02	Cholesterol Level	mg/dl
03	Systolic Blood Pressure (SBP)	mmHg
04	Diastolic Blood Pressure (DBP)	mmHg
05	Heart Rate	BPM
06	Age	days
07	Height	cm
08	Weight	Kg
09	Gender	0 = Male; 1 = Female
10	Smoking habit	0 = No; 1 = Yes
11	Alcohol intake	0 = No; 1 = Yes
12	Physical activity	0 = Inactive; 1 = Active

Equations

$$\text{Glucose (mg/dL)} = -0.0011X + 194.87 \dots\dots\dots(i)$$

$$\text{Cholesterol (mg/dL)} = -0.0033X + 531.49 \dots\dots\dots(ii)$$

here, X is the average value of IR

$$\text{SBP (mmHg)} = 184.3 - 1.329 \times \text{HR(bpm)} + 0.00848 \times \text{Td} \dots\dots(iii)$$

$$\text{DBP (mmHg)} = 55.96 - 0.02912 \times \text{HR(bpm)} + 0.02302 \times \text{Td} \dots\dots(iv)$$

here, $\text{Td} = \text{HR(ms)} - \text{Time Delay}$; $\text{Time Delay} = \text{HR(bpm)} + \text{PTT}$

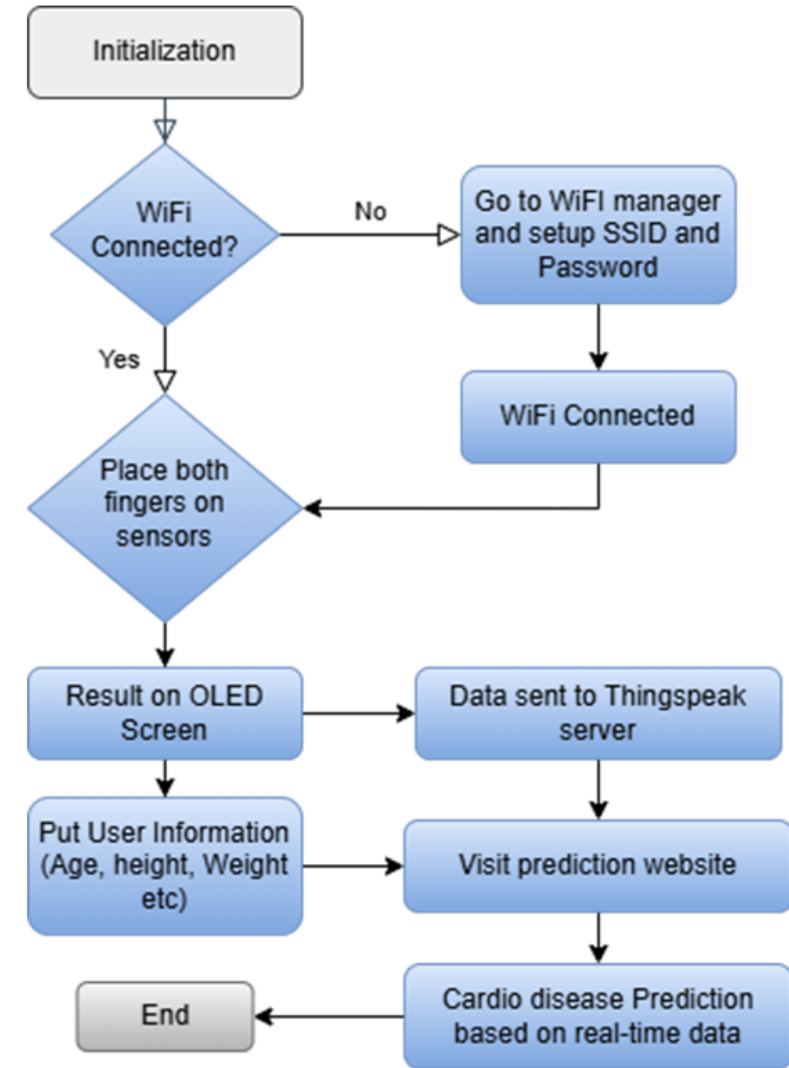


Fig. 7: Flow chart of the CardioPredictor system

Results



Fig. 8: Measured data on OLED screen



Fig. 9: Measured SBP and DBP on arm blood pressure device

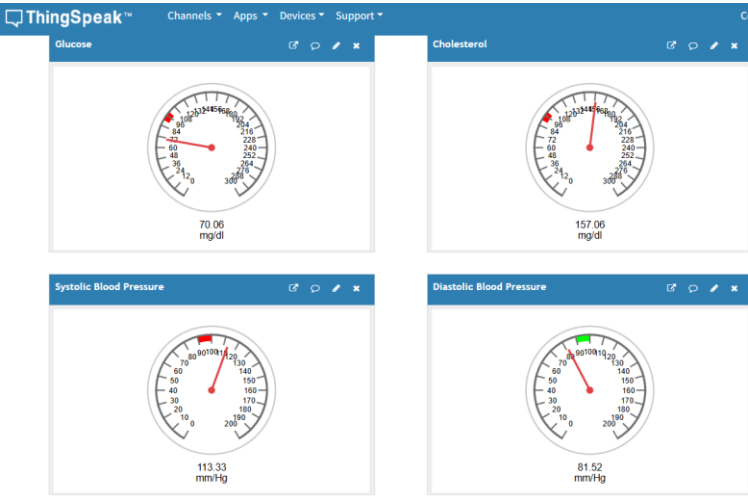


Fig. 10: Data sent to ThingSpeak Server

Heart Disease Prediction Result

There is a 17.00% chance of having cardiovascular disease.

Sensor Data Used:

Cholesterol: 157.06 mg/dL

Glucose: 70.06 mg/dL

Systolic Blood Pressure (SBP): 113.33 mmHg

Diastolic Blood Pressure (DBP): 81.52 mmHg

Make another prediction

Fig. 12 Predicted result based on realtime data

Heart Disease Prediction System

Age (years):

24

Gender:

Male

Height (inches):

66

Weight (kg):

70

Smoking:

No

Alcohol intake:

No

Physical activity:

No

Glucose (mg/dL):

70.06

Cholesterol (mg/dL):

157.06

Systolic BP (ap_hi):

113.33

Diastolic BP (ap_lo):

81.52

Predict

Fig. 11: Data fetched into website

Validation

Table 2: Comparison between CardioPredictor and Accu Answer blood glucose measuring device

Statistic	Cardio-Predictor (Avg)	Accu Answer (Avg)	Offset Correction (mg/dL)	Variation (%)
Mean	110.5	118.2	-7.7	6.55
Standard Deviation	10.2	11.5	-0.3	2.61
Min Value	70	77	-8.9	8.25
Max Value	160	169	-7.4	7.10

Table 3: Comparison between CardioPredictor and Digital Blood pressure measuring device

Statistic	SBP – Cardio-Predictor (mmHg)	SBP – Digital (mmHg)	DBP – Cardio-Predictor (mmHg)	DBP – Digital (mmHg)
Mean	117.29	125.35	79.69	85.85
Standard Deviation	11.09	16.67	8.04	8.33
Min Value	96.48	101.45	64.65	68.70
Max Value	142.23	151.30	99.71	105.18

Model Accuracy

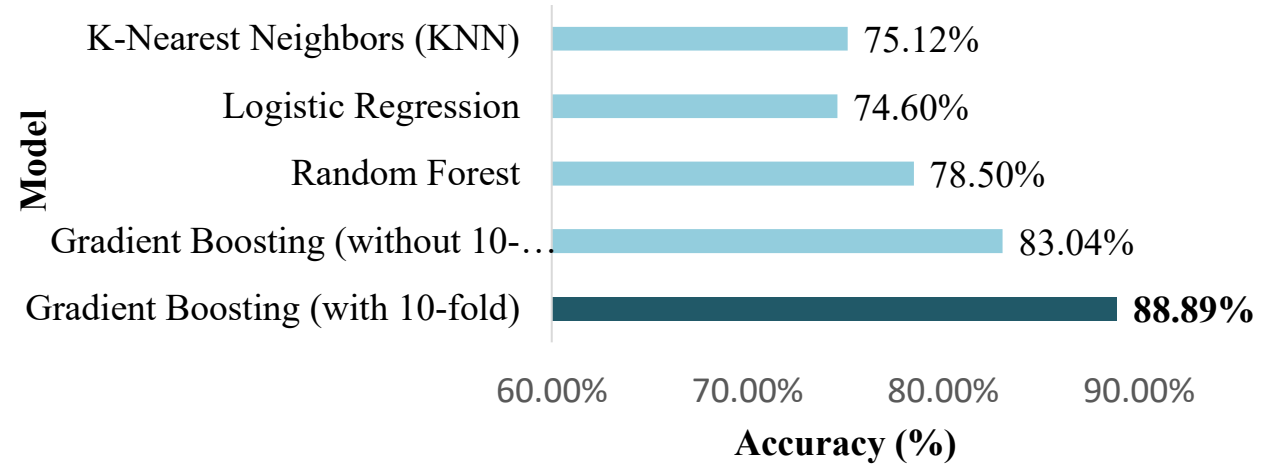


Table 14: Comparison between CardioPredictor and Accu Answer Blood Cholesterol measuring device

Statistic	Cardio-Predictor (Avg)	Accu Answer (Avg)	Offset Correction (mg/dL)	Variation (%)
Mean	200.3	215.6	-15.3	7.38
Standard Deviation	14.6	16.3	-0.7	3.89
Min Value	183.0	200.2	-17.2	8.58
Max Value	219.4	226.3	-14.9	6.76